

INTRODUCTION TO COMPUTER SYSTEM, NETWORKS AND SECURITY

Assignment 2 part 1 – MARIE Programming (Reflection)

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Purpose for the whole assignment 2:

- ❖ This assignment is mainly about programming in the MARIE assembly language.
- ❖ This will allow students to demonstrate their comprehension of the fundamental ways a processor works.
- ❖ In addition, basic knowledge about processes, memory and I/O is assessed in an in-class interview.
- ❖ The assignment relates to Unit Learning Outcomes 2, 3 and 4.

Purpose for the assignment 2 part 1(Reflection):

TASK	BRIEF DESCRIPTION
Reflection	Reflect on what was done during week 4, week 5 and week 6 of classes; what you learned, what was difficult...

FIT1047 Weekly Reflection

Week 4:- Machine code, CPUs and memory

Key Learnings:

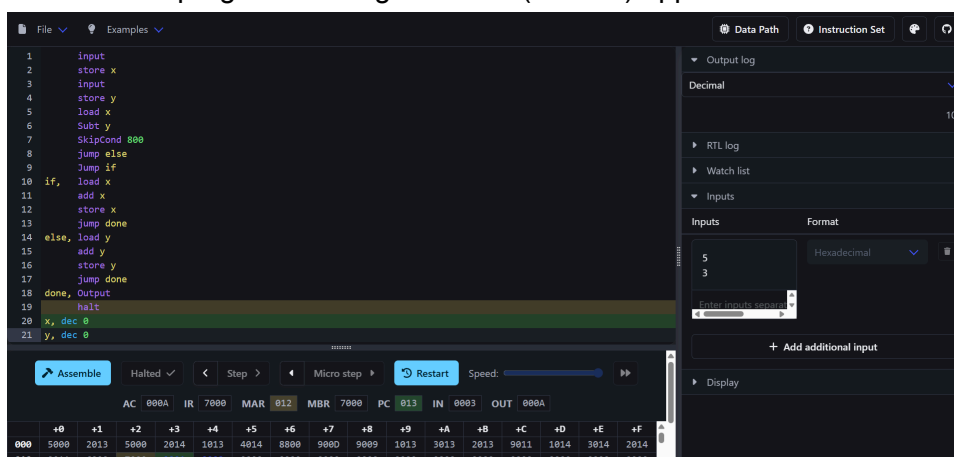
- I was already familiar with the basic assembly operand(opcode){albeit only on paper, not any any software/hardware) as it was a learning requirement for my A-levels. What was new to me was
 - Actually being able to run the assembly code on MARIE, seeing the values in memory location change.
 - The following opcode was also new to me;
 - ORG
 - SkipCond
 - JnS
 - ADR
 - Subt

Personal Insights:

- What I found fascinating is that MARIE is a small scale version of a pc or laptop that we are using, and my actual laptop is doing all that that marie is doing but on a much larger scale.
- Also the shire amount of line of code to do simple task is mesmerising:
 - For example; Part 3 task 3, the whole program is around 1100 lines of MARIE code, if the same program were to be done in high level language(e.g python), it would take a maximum of 100 lines of code.

Application of Knowledge:

- Coded some programs during this week(week 4) applied class



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Challenges Faced:

- The biggest challenge I faced was getting used to using Skipcond as it can't compare data directly, some prior code is needed to do the comparison.

Future Goals:

- I am definitely getting used to using MARIE, so i will continue to code to better myself.

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Week 5:- Operating Systems, Input/Output and the boot process

Key Learnings:

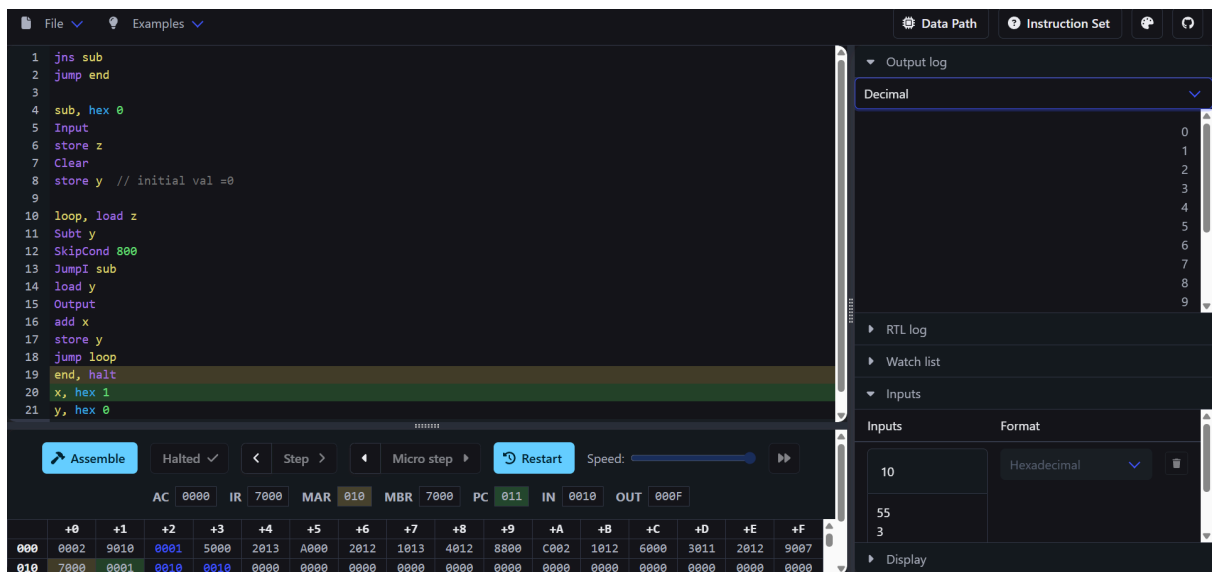
- The subroutines; I have never written/coded any subroutine for assembly. This was new to me so, I struggled a little bit with my first through program but fortunately got used fast and now I can code and run subroutines easily.
- Learning about how to display things was also very important.

Personal Insights:

- I noticed how difficult it is to output things on the display as each pixel of what is to be output needs to be changed, for example displaying character A requires a lot of lines of code.

Application of Knowledge:

- I applied this knowledge during applied class for week 5

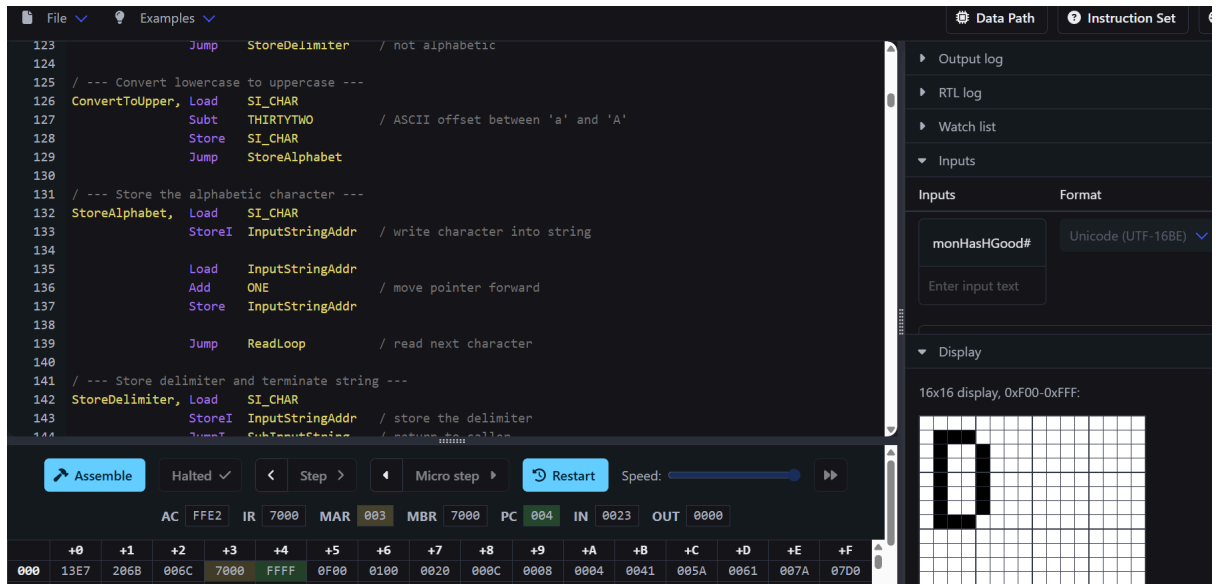


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- I also used that knowledge to do assignment 2 part 3



Challenges Faced:

- When disassembling, I have difficulties differentiating between whether it is JnS or just a data value whenever there is 0XXX (where X is some hexadecimal value)
- Another challenge I face is how to use the display. It was a little tricky to use it at first but after messing around with it I became better at using it.

Future Goals:

- I am now confident in my knowledge of the assembly language. I continue to code programs on MARIE to keep my knowledge fresh.
- I will also start reading the materials for next week to get a head start on the topic.

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Week 6:- Networks: layer model, application layer

Key Learnings:

- Understood how data flows between CPU, memory, and I/O devices using memory-mapped I/O vs port-mapped I/O.
- Learned that application code cannot directly access hardware; instead, it requests services through the OS using system calls.
- Explored multitasking concepts (e.g., round robin scheduling) and how time slices affect turnaround time.
- Differentiated between **virtual memory** and **physical memory**, and how address mapping allows multiple processes to coexist safely.
- Practiced MARIE programming to handle input, output, and character manipulation using subroutines.

Personal Insights:

- Realized how crucial the OS is as a “manager” that balances resources, ensures fairness, and protects hardware access.
- Saw how scheduling policies impact performance; a small change in time quantum can alter turnaround time significantly.
- Appreciated that concepts like memory mapping and process scheduling aren’t abstract, they directly affect real-world performance of applications.

Application of Knowledge:

- I can now explain how user applications rely on system calls to request hardware services.
- Able to write simple MARIE programs that demonstrate I/O handling and character conversion, showing the OS-like role in managing resources.
- Can apply round robin scheduling calculations to predict performance outcomes in different scenarios.

Challenges Faced:

- Understanding how virtual addresses map to physical memory was initially confusing, especially base and bounds registers.

Future Goals:

- Strengthen understanding of more advanced scheduling algorithms (e.g., priority scheduling, shortest job next).
- Explore how these OS concepts scale to modern multi-core systems(like my laptop).